DATA ANALYSIS LAB · SECTION

Data **Visualization**

`iris_dataset · 150 rows · 5 columns`

📋 SYNTAX CHEATSHEET

New canvas	`plt.figure(figsize=(8, 5))`	Display the chart	`plt.show()`
Histogram	`plt.hist(df['col'], bins=20)`	Chart title	`plt.title('text')`
Boxplot	`plt.boxplot(data, labels=names)`	X-axis label	`plt.xlabel('text')`
Scatter plot	`plt.scatter(df['x'], df['y'])`	Y-axis label	`plt.ylabel('text')`
Vertical line	`plt.axvline(value, color='red')`	Add legend	`plt.legend()`
Dashed line style	`linestyle='--'`	Fix spacing	`plt.tight_layout()`
Mean of column	`df['col'].mean()`	Median of column	`df['col'].median()`

▶ RUN THIS FIRST IN YOUR NOTEBOOK — DO NOT CHANGE IT

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

iris_raw = load_iris()
df = pd.DataFrame(iris_raw.data, columns=['sepal_length', 'sepal_width',
'petal_length', 'petal_width'])
df['species'] = iris_raw.target_names[iris_raw.target]

# Columns: sepal_length | sepal_width | petal_length | petal_width | species
```

QUESTIONS

Q1 HISTOGRAM

Plot a histogram of `petal_length` with **20 bins**.

Add a **red vertical line** at the mean and a **green dashed vertical line** at the median.

Add a title, x-axis label, y-axis label, and a legend.

Hint: Calculate mean and median before drawing. Use label= inside axvline to make the legend work.

Q2 BOXPLOT

Plot a boxplot comparing `petal_length` across the **3 species** side by side.

Make the **median line red**. Add a title, x-axis label, and y-axis label.

Hint: You need to separate the data into 3 groups first — one list of values per species.

Q3 SCATTER PLOT

Plot a scatter plot of `petal_length` (x-axis) vs `petal_width` (y-axis).

Color each species **differently**. Add a title, axis labels, and a legend.

Hint: Loop through species using groupby. Use a colors dictionary to assign a color per species.

Q4 ⭐ BONUS

Plot a histogram of `sepal_length` instead of `petal_length`.

Use **15 bins** and add the mean and median lines. What do you notice compared to Q1?